

Comparison of Programmable FPGA Technologies

FPGA	Advantages	Disadvantages
Antifuse-based FPGAs	Very low power consumption, instant-'on' when powered up, radiation-tolerant, highly secure (which makes it hard to reverse engineer the design)	One-time programmable
Flash-based FPGAs	Low power consumption, instant-'on,' radiation-tolerant, reprogrammable	Additional fabrication required on top of the standard CMOS process, low gate count
SRAM-based FPGAs	Based on the standard CMOS process, reprogrammable, available in large gate counts	Volatile (reconfiguration required on power-up), higher power consumption

velopment kit for Altera's Cyclone III FPGAs provides soft-core implementation. Xilinx' Microblaze, Altera's Nios II, and Lattice Mico 8 and Mico 32 are also soft-core implementations. The concept of multi-core processors and parallel computing can be extended to FPGAs in order to perform more complex embedded tasks.

New concept: PSoC

Cypress Semiconductors has introduced a new concept of programmable system-on-a-chip (PSoC). Rajeev Mehtani, senior vice president, Cypress Semiconductor, explains, "PSoC devices encapsulate features of a microcontroller with flexibility and programmability of FPGAs and a configurable analogue sub-system." "Our chip comes with a sophisticated software that allows the users to program the digital and analogue sub-systems to their specific application, giving them ultimate control of the system design process," he adds.

Scope for improvement

"FPGA as a technology has evolved, and has started developing a lot of ecosystem around it," says Rahul V. Shah, director, custom solutions, elfnchips. "The ecosystem means FPGAs with different features, different IP cores, tools, evaluation board, reference design from the vendor and ultimately vendor support." With all this, time-to-market is another challenge before the vendors and designers.

FPGA vendors are providing newer FPGA chips with more and more func-

tionality in sync with the market demand. However, in the race, the chips are not tested and verified thoroughly. "There is no chip which is bug-free" has now become applicable to FPGAs, which are coming with added complexities," says Shah. If the new chip is a derivative device, i.e., a variant of the already existing FPGA chip, it has the least probability of having a bug.

There are still certain application areas where ultra-low-power ASIC chips are preferred to FPGAs; e.g., in handheld devices. "FPGA technology has not reached a point so far where it will find its application in ultra-low-power devices. However, for a few handheld devices where performance is needed, such as the man-pack radio carried by soldiers for highly sophisticated wireless communication, FPGAs are preferred," shares Varma.

FPGAs sometimes show erratic behaviour beyond certain temperature. "You cannot select an FPGA for your thermal requirement with just one parameter. You have to do a comparative study to determine whether the FPGA works fine at a particular temperature or at a particular frequency on a specific board," comments Shah.

Aggarwal adds, "As long as you are working with the vendor's board, the specifications are well-mentioned. But for a custom-made board, these are not well documented." The vendor documentation does not give you the exact details of the required cooling fan, kind of cooling, temperature and even the weight of the fan—things that affect the device design.

Software tools for FPGA are still a problem for designers. Aggarwal says, "The tools are locked to a particular vendor and designers need to pay for the FPGA device as well as tools to the vendor."

Prabhakar adds, "Every FPGA has its proprietary tool, so changing FPGAs in subsequent projects requires learning of a new design tool suite altogether." There is a need for platform-independent tools for FPGA designers.

Despite technological advancement, FPGAs are still high on bill-of-material (BOM) price. "In most cases, the most expensive chip in a typical board BOM comes out to be an FPGA," says Prabhakar.

In a nutshell

FPGAs have moved from 130nm to 28nm process today. There is a lot of development and innovation happening to support additional features, programmability and flexibility. Also, the industry is extending support in terms of training the users on basic architecture, providing reference designs and helping them in custom designs.

"The bulk of the revenue for FPGAs comes from wireless and wired communications. This is the fastest growing segment across the globe. Display and security and surveillance are the other fast-growing applications based on FPGAs. These are the areas where we see India as a huge market," informs Peera.

"FPGAs are the best choice when volumes are of the order of a few hundreds per annum. These are also the best choice while designing prototypes or proof-of-concept products as well as stepping stone towards ASIC design. All this is possible as FPGAs are reprogrammable, field-upgradable and easy to design. With all the new developments, FPGAs are preferred in applications that are complex and performance-intensive," says Prabhakar. ●

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